

Literature Status Note for arXiv:2010.13098

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Status

This is a lightweight literature-status packet. The dual-level half of Question 1 in Candido–Kaufmann, “On the geometry of Banach spaces of the form $\text{Lip}_0(C(K))$ ”, arXiv:2010.13098, is already answered by a later paper: Candido–Guzman, “On large ℓ_1 -sums of Lipschitz-free spaces and applications”, arXiv:2202.09932.

Source Question

The source paper asks in Section 3, p. 3, Question 1: for an infinite Hausdorff compactum K with $w(K) = \kappa$, under which hypotheses can one guarantee

$$\mathcal{F}(C(K)) \sim \mathcal{F}(c_0(\kappa)),$$

or at least the dual-level classification

$$\text{Lip}_0(C(K)) \sim \text{Lip}_0(c_0(\kappa)).$$

The same paper records this as only partially answered at the dual level by its Corollary 4 and Theorem 5.

Later Answer

Candido–Guzman, arXiv:2202.09932, state in the introduction, p. 2, Theorem 3: if X is an $\mathcal{L}_p$ space of density κ , then

$$\text{Lip}_0(X) \sim \begin{cases} \text{Lip}_0(\ell_p(\kappa)), & p < \infty, \\ \text{Lip}_0(c_0(\kappa)), & p = \infty. \end{cases}$$

They then state Corollary 4: if K is an infinite Hausdorff compact space of weight κ , then

$$\text{Lip}_0(C(K)) \sim \text{Lip}_0(c_0(\kappa)).$$

Why This Matches

Every Banach space $C(K)$ is an $\mathcal{L}_\infty$ space, and for infinite Hausdorff compact K one has $\text{dens}(C(K)) = w(K)$. Applying the $p = \infty$ case of Candido–Guzman’s Theorem 3 to $X = C(K)$ gives exactly the dual-level alternative asked for in Question 1.

The supporting paper explicitly knows this relation: it cites the Candido–Kaufmann paper as reference [5] and says its Corollary 4 extends the main result of [5] in the dual setting. Therefore this belongs in `solutions/literature_already_answered/`, not in the original-results folders.

Limitations

This packet does not answer the free-space part of Question 1, namely whether

$$\mathcal{F}(C(K)) \sim \mathcal{F}(c_0(\kappa))$$

for all infinite Hausdorff compacta K of weight κ . Only the “at least Lip_0 ” half is cleared.

References

- L. Candido and P. L. Kaufmann, “On the geometry of Banach spaces of the form $\text{Lip}_0(C(K))$ ”, arXiv:2010.13098; Proc. Amer. Math. Soc. 149 (2021), 3335–3345.
- L. Candido and H. H. T. Guzman, “On large ℓ_1 -sums of Lipschitz-free spaces and applications”, arXiv:2202.09932; Proc. Amer. Math. Soc. 151 (2023), 1135–1145.